

M40 — Pre-populate Room of N Characters + Curriculum drawers from Activity

Phase: Phase 7 follow-on (post-M39, operator-feedback fix) **Status:** **DONE** **Blocked by:** M12 (Activity catalog + per-row picker), M31 (per-row Characters/Curriculum drawers) **Blocks:** none **Estimated effort:** 0.5 day

1. Purpose

Operator feedback after M39:

"For Room of N for the encounters the characters and curriculum should pre-populate like they do for the single character."

In the single-patient wizard, picking a sample at Step 2 calls `applySample(s)` which:

- Sets `scenario_name`, `scenario_notes`, `scenario_text`.
- Sets the program + week selects + populates the week-derived module list.
- Checks every persona checkbox in `s.personas`.
- Checks every module checkbox in `s.modules`.

In Room of N mode (M31), picking an Activity on a row already pre-filled the scenario textarea + label + primary persona dropdown and stashed `seed_modules` into a dataset for submit. But the **drawer checkboxes stayed empty** — the operator had to manually open the Characters drawer and check the persona, open the Curriculum drawer and check every module. Even though the seed data was there, the UI didn't surface it.

M40 mirrors `applySample`'s drawer-population step for each row. Plus two related fixes:

1. When the row's primary persona dropdown changes, auto-check the matching Characters-drawer checkbox (the "primary is always part of the cast" invariant the submit logic enforced at the end — now surfaced earlier in the UI).
2. When the operator bumps the room-N input, `renderRoomEncounterRows` re-renders every row from the captured `prev` array. M40 extends `prev` to also capture each row's drawer state (`personaList`, `modulesList`, `programId`, `week`) so re-renders don't wipe the operator's selections.

2. Structure

Files touched:

- `portal/static/control.js`:
- Extended the Activity-change handler (``document.addEventListener`

(`"change", ...)` for `[data-field="activity"]`) — after stashing seed data and pre-filling the textarea, programmatically check `[data-row-persona][value=<seed_persona_id>]` and every `[data-row-module][value=<each seed_module>]`. Refresh tab-strip badges via the new helper.

- New `cssEscape(s)` polyfill — defensively quotes persona/module ids for attribute selectors. Falls through to `CSS.escape` when available.
- New `updateRowTabBadges(row)` helper — reads checkbox state, writes both badge counts. Single source of truth for row badge arithmetic.
- New persona-dropdown change handler — when `[data-field="persona"]` changes on a row, auto-check the matching Characters-drawer checkbox and refresh badges.
- `renderRoomEncounterRows` `prev` capture — now reads

personaList, modulesList, programId, week from each row before re-rendering. The existing render code already consumed existing.personaList / existing.modulesList / existing.programId / existing.week`; this fills those reads with the operator's actual prior state.

No backend change. No HTML/CSS change. No new dataclass field.

3. Uses

3.1 Operator flow

1. Operator goes through wizard → Room of N → Step 4r (Encounters). 2. Each row defaults: empty drawers, primary persona dropdown set to the first persona in the catalog. 3. Operator picks an Activity on row 1 — say `"ED Sepsis"` with `seed_persona_id = "P-014"` and `seed_modules = ["M32", "M08", "M02"]`. 4. **Old behavior:** scenario textarea + label + primary persona dropdown updated; drawer checkboxes stayed empty; submitting sent the row with only the primary persona + (via dataset fallback) the seed modules. Operator had no way to know seed modules would land — had to trust the dataset. 5. **New behavior (M40):** same updates PLUS: - Characters drawer's P-014 checkbox is now checked. - Curriculum drawer's M32, M08, M02 checkboxes are all checked. - Badge strip on the row updates: Characters · 1, Curriculum · 3. - Operator can immediately open either drawer and see exactly what will be sent — and add or uncheck as needed. 6. If the operator picks a **different** primary persona via the row's persona dropdown (after the Activity pick), the new persona's checkbox in the Characters drawer auto-checks too (without unchecking the previous one — multi-cast intent). 7. If the operator bumps the room-N input from 4 to 6, the four existing rows re-render preserving all their drawer state. The two new rows render empty.

3.2 Why mirror applySample semantics exactly

Operators move between single-patient and Room of N modes within the same session. Carrying the same mental model — "picking a template fills the persona + module checkboxes" — across both modes removes a class of confused-operator bug reports. The behavior should be:

Single-patient pick a sample	Room of N pick an Activity
Fill scenario_name + notes	Fill row's label (M12)
Check personas	Check row's Characters drawer (M40)
Check modules	Check row's Curriculum drawer (M40)
Set program + week	(Activity has no program/week — left blank)

The "set program + week" row is unfilled in Room of N because the Activity dataclass (M11) doesn't currently carry `program_id` / `week`. Out of scope for M40 — the operator can still set them per row if they want, and the Step 3-equivalent is per-row drawer in Room of N (since M32 hid the wizard-wide Step 3).

4. Functions (exported API surface)

Symbol	Where	Purpose
<code>updateRowTabBadges(row)</code> (new)	<code>portal/static/control.js</code>	Reads <code>[data-row-persona]:checked</code> and <code>[data-row-module]:checked</code> counts and writes both badge spans. Single source of truth.
<code>cssEscape(s)</code> (new)	<code>portal/static/control.js</code>	Defensive polyfill for <code>CSS.escape</code> — used to safely embed persona/module ids in attribute selectors.

Symbol	Where	Purpose
(extended) Activity-change handler	same	After stashing seed data, checks the matching persona + module checkboxes and refreshes badges.
(new) Persona-dropdown change handler	same	When the row's primary persona dropdown changes, auto-checks the matching Characters-drawer checkbox.
(extended) renderRoomEncounterRows	same	prev capture now includes personaList, modulesList, programId, week.

5. Limitations

- **Activity seed data has no program/week.** The Activity

dataclass (M11) carries seed_persona_id (single primary) + seed_modules (list) + scenario_text. To also pre-fill the Curriculum drawer's program + week, a future M41 would need to extend the catalog. Out of scope today — operators can set per- row program + week directly in the drawer.

- **seed_persona_id is one persona only.** Real scenarios often

involve a cast (patient + family + MD + RN). The Activity model reflects an MVP. A future M41 could carry seed_personas: [pid, pid, ...]; M40 would happily check all of them.

- ****Re-picking an Activity does NOT uncheck the prior Activity's**

modules**. By design — additive seed sets feel less surprising than wiping the drawer. Operators who want a clean slate can uncheck manually or pick "— (no activity) —" first.

- **The persona-dropdown change handler is additive too.** Picking

a new primary doesn't uncheck the prior primary's Characters- drawer entry. Matches the "cast" mental model — same rationale as the Activity additivity above.

- **No keyboard shortcut to open/collapse all drawers.** Operators

click row-by-row. Acceptable; the 10-encounter cap (M19) keeps it manageable.

- ****Drawer state captured in prev is keyed by row position, not**

row id**. If the operator drags a row to reorder (no UI for that today), the captured state stays at the old position. N/A today; document if reordering ships.

6. Test status

Test file	Asserts	Status	Last run
tests/v7/test_room_row_prepopulate.py::test_activity_handler_checks_seed_persona_checkbox	Activity handler checks [data-row-persona] matching a.seed_persona_id	PASS	2026-05-27
...::test_activity_handler_checks_each_seed_module_checkbox	Handler iterates [data-row-module] and checks each in Set(a.seed_modules)	PASS	2026-05-27

Test file	Asserts	Status	Last run
...::test_activity_handler_refreshes_badge_counts	Handler calls updateRowTabBadges(row) after the auto-check	PASS	2026-05-27
...::test_persona_dropdown_change_auto_checks_characters_drawer	Primary-persona dropdown change handler auto-checks the matching Characters-drawer checkbox + refreshes badges	PASS	2026-05-27
...::test_updateRowTabBadges_helper_exists	Shared helper exists and references both data-row-* selectors + .row-tab-count	PASS	2026-05-27
...::test_prev_capture_includes_drawer_state	prev array captures personaList, modulesList, programId, week	PASS	2026-05-27
...::test_row_render_uses_existing_persona_and_module_lists	Row template still consumes existing .personaList / existing.modulesList (M31's wiring is intact)	PASS	2026-05-27
...::test_cssEscape_polyfill_exists	Defensive CSS.escape polyfill is present	PASS	2026-05-27
Full v7 suite	263 passed, 1 skipped (M20 Playwright skip — unchanged)	PASS	2026-05-27

7. Change list

Date	Author	Change	Files
2026-05-27	claude-code	Initial M40 implementation: Activity-handler auto-check for persona + modules; persona-dropdown change handler; updateRowTabBadges + cssEscape helpers; prev capture extended for drawer state; 8 source-guard tests	portal/static/control.js, tests/v7/test_room_row_prepopulate.py (new)

8. Open questions / known issues

- Should the Activity catalog carry program + week? Single

scenarios already pre-fill those at Step 2. Extending the `Activity` dataclass with `program_id + week` would close the parity gap. Tracked for M41.

- ****Should picking a *new* Activity uncheck the prior Activity's**

modules?** M40 chose additive (don't surprise operators with unexplained un-checks). If LAN test feedback says the additive behavior feels wrong, flip to a "clear and re-fill" pattern.

- **The cast model.** Today the catalog has `seed_persona_id`

(single). Real wedge characters (e.g. "Anxious Spouse", "Charge RN") exist in the persona library but aren't surfaced via the catalog. A `seed_personas: list[str]` field on `Activity` would let scenarios prescribe a full cast.

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